

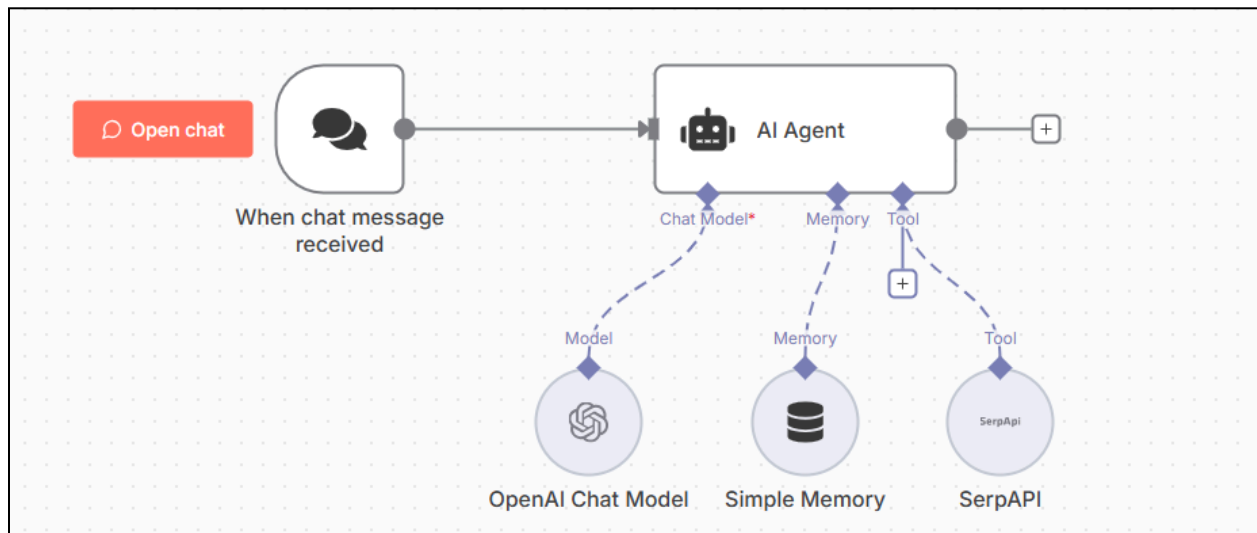
Overview

The n8n workflow gets a real-time, AI-generated summary of the latest news about US tariffs on India, along with a table showing sectoral investment impacts. The flow is initiated when a user sends a chat message (natural language input via chat). It then searches for recent news, processes it through a memory component, summarizes using an AI model, and delivers the final response through the AI Agent node.

The goal is to:

1. Let the user send a question like "What are the latest US tariffs on India?"
2. Search the news using SerpAPI
3. Summarize the news using OpenAI
4. Show a sector-wise impact table
5. Send everything back to the user via the AI Agent

Workflow:



Workflow Components

Here are the main components used in this workflow:

Node	Purpose
Chat Message Trigger	Starts the workflow when the user sends a chat message.
AI Agent	Central brain: sends prompt, calls tools, builds response.
SerpAPI	Finds the latest news articles on the topic.
Simple Memory	Stores and reuses the user query or intermediate data.
OpenAI Chat Model	Used by the AI Agent to summarize content and generate final output.

Step-by-Step Guide to Build the Workflow

Follow these steps inside your n8n editor:

Step 1: Add the Chat Message Trigger

- This node initiates the workflow when a user sends a message.
- In n8n, drag in a Chat Message Trigger node.
- Set:
 - Trigger Type: Chat Message
 - Example input:
" Summarize the US tariffs on India and how they affect different investment sectors. "
- Save and name the node as "Start - Chat Trigger"

Step 2: Add the AI Agent Node

- This is the most important node. It will:
 - Read the user's input
 - Decide what tools (like SerpAPI, OpenAI, and Memory) to use
 - Host the response for the user
- Add an AI Agent node and connect it to the Chat Trigger.
- In the AI Agent node, set:
 - Source for Prompt (User Message) as "Connected Chat Trigger Node"
 - In Options, add System Message:

"

You are a helpful AI.

Search for the latest news, government notifications, and expert analyses on the recently declared US tariff structure on India. Summarize the findings with a focus on investment implications.

Instructions:

Extract the industry domain from the user query.

Identify impacted sectors – List which sectors are directly affected by the new tariffs. Explain how tariffs will impact each sector (exports, imports, pricing, demand, supply chain).

Highlight indirect impacts – Note sectors that could be indirectly affected due to supply chain disruptions, input cost changes, or market sentiment.

Investor perspective – Clearly outline:

Sectors to avoid in the short-term (due to volatility, trade barriers, or reduced competitiveness).

Sectors likely to remain resilient or benefit despite tariffs (e.g., those with strong domestic demand, alternative export markets, or government support).

Provide context – If available, include expert commentary, trade data, or policy notes to support the analysis.

Output:

Deliver output in a structured format –

Directly Impacted Sectors

Indirectly Impacted Sectors

Sectors to Avoid (Short-Term)

Resilient / Beneficiary Sectors

Make sure that the news extracted is strictly aligned with the "US tariffs on India" domain.

Generate the output in the form of a structured table.

“

Step 3: Add the SerpAPI Node

- This fetches the most recent Google News results.
- Add a **SerpAPI** node
- In the parameters:
 - Credential to connect with: SerpAPI account
 - In case the SerpAPI account does not show as an option, please create a SerpAPI account and configure the API Key

Step 4: Add the Simple Memory Node

- This node helps store and access the user's original question across the conversation.
- Add a **Simple Memory** node. Set:
 - Session ID: Connected Chat Trigger Node
 - Context Window Length: 5
- Again, this is not manually connected to any other nodes — the AI Agent will use it as a tool.

Step 5: Add the OpenAI Chat Model Node

- This node is used by the AI Agent to generate a summary and output table.
- Add an **OpenAI Chat Model** node. Set:

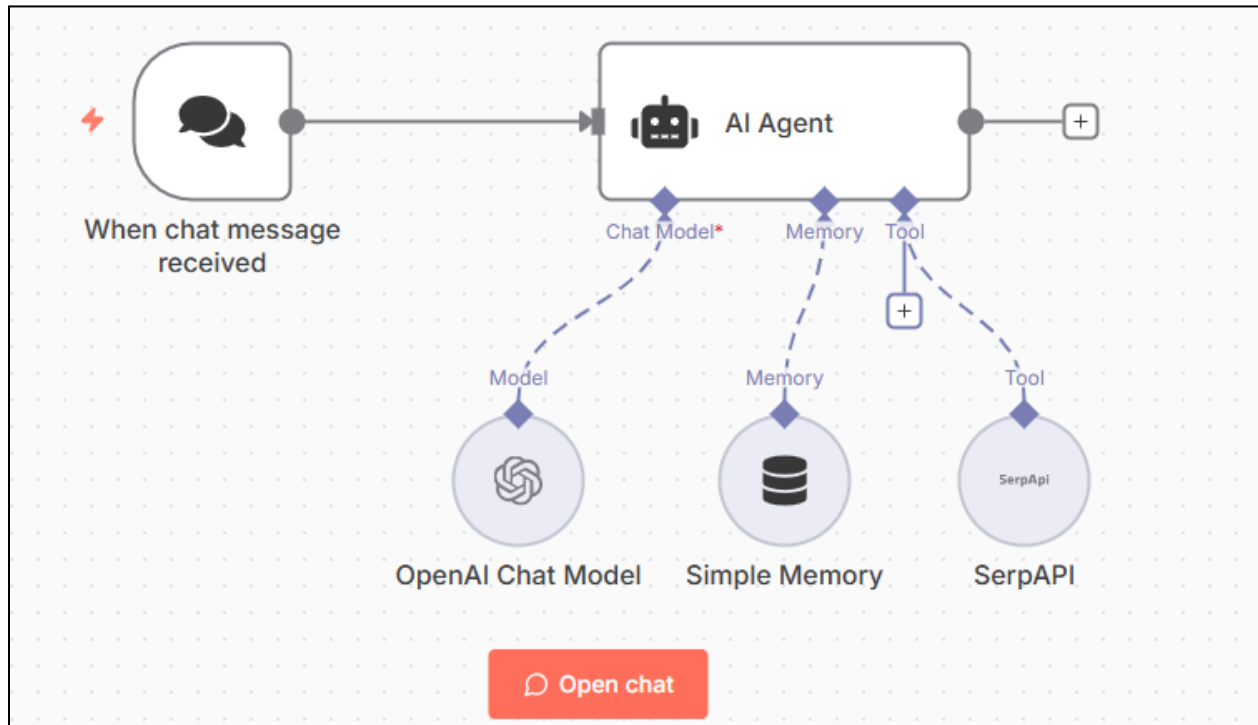
- Credential to connect with: n8n free OpenAI API credits
(or add OpenAI account details)
- Model:
From list: gpt-4o-mini

The workflow is all set!

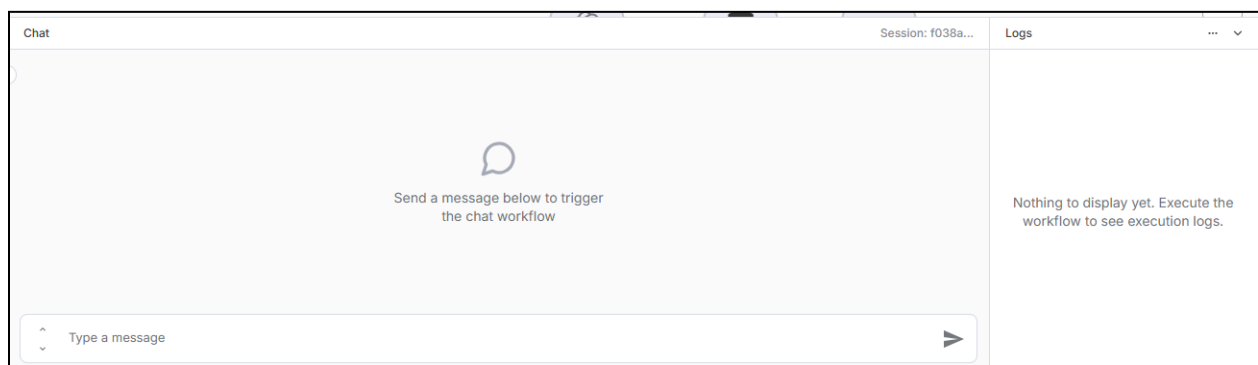
Test the Workflow

Once the workflow is set, we can test the same by following the steps below:

Step 1: Click on “Open chat” button.



Step 2: An interface (as in screenshot below) will pop-up. Add a prompt here to trigger the workflow.



Step 3: Wait for the workflow to execute. Once done, you shall be able to see the output generated.

The screenshot displays a workflow execution environment. At the top, a workflow diagram shows a sequence: 'When chat message received' (1 item) → 'AI Agent' (1 item) → 'Output' (1 item). The 'AI Agent' node is connected to three sub-nodes: 'OpenAI Chat Model' (4 items), 'Simple Memory' (2 items), and 'SerpAPI' (4 items). Below the diagram, the 'Chat' panel shows a session with the prompt: 'Summarize the US tariffs on India and how they affect different investment sectors'. The 'Logs' panel shows the execution steps: 'When chat message received', 'AI Agent', 'Simple Memory', 'OpenAI Chat Model', 'SerpAPI', 'OpenAI Chat Model', and 'SerpAPI'. The 'Output' panel shows the generated response: 'Based on the recent analysis of the U.S. tariffs on India, here is a summarized table of the findings, with a focus on the investment implications:'. The response is formatted as a table with columns 'Category' and 'Details'. The table lists 'Directly Impacted Sectors' and 'Textiles' as categories, with details about increased costs, demand reduction, and higher vehicle prices. The response also lists 'Auto Parts' and 'Agriculture' as categories, with details about increased input costs and reduced exports.

Category	Details
Directly Impacted Sectors	Textiles: Higher tariffs may lead to increased costs, affecting price competitiveness in the U.S. market.
	 Gems and Jewelry: Increased tariffs will likely reduce demand due to higher consumer prices.
	 Auto Parts: U.S. manufacturers may face increased input costs, leading to higher vehicle prices.
	 Agriculture: Specific products (ex. certain spices) may experience reduced exports due to tariffs, affecting

...economic activities.